

DRAWING LIST FOR TEST PACKAGE**WA-632****PROJECT NO.: 22328D****PROJECT TITLE : M15X Ph2 Project**

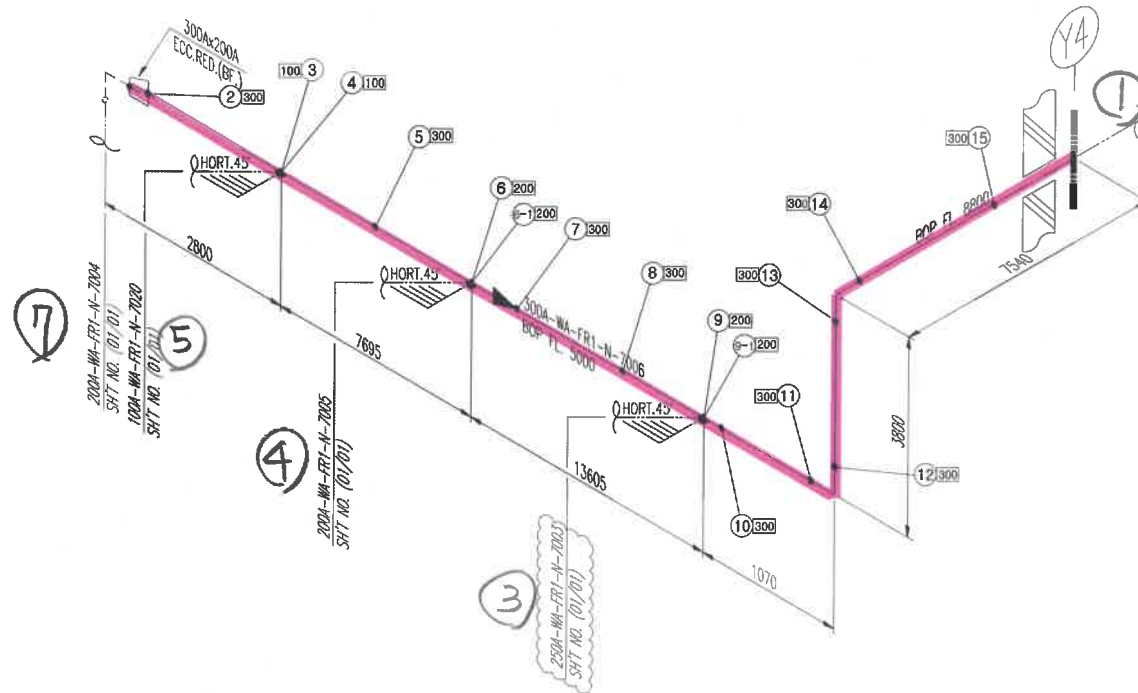
No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	632	2FA	300A-WA-FR1-N-7006-01	1 OF 1	0	CJ-M15W14-PR-A20-000167	2A	
2	632	R2FA	300A-WA-FR1-N-7006-01	1 OF 1	0	CJ-M15W14-PR-A20-000501.502	1A	
3	632	R2FA	250A-WA-FR1-N-7003-01	1 OF 1	0	CJ-M15W14-PR-A20-000501	1A	
4	632	R2FA	200A-WA-FR1-N-7005-01	1 OF 1	0	CJ-M15W14-PR-A20-000501,503	1A	
5	632	R2FA	100A-WA-FR1-N-7020-01	1 OF 1	1	CJ-M15W14-PR-A20-000504	1A	
6	632	R1MF	100A-WA-FR1-N-7020-01	1 OF 1	0	CJ-M15W14-PR-A20-000504	1A	
7	632	R2FA	200A-WA-FR1-N-7004-01	1 OF 1	0	CJ-M15W14-PR-A20-000502,503	1A	




NOTE

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1A	REVISED AFC	25.00.14
0	REVISED AFC	25.01.13
1	APPROVE FOR CONSTRUCTION	24.10.31
수령 REV	설명 DESCRIPTION	날짜 DATE
제 DIAGRAM	이 이입학	
결 결	한 한차례	
시 시	입 입	
APPROVE FOR	바 바	
제 제	작 작	
SK 에코플랜트		
공사명 PROJECT TITLE		
M15X Ph4 WWT Project		
도면명 DWG. TITLE		
PIPING & INSTRUMENT DIAGRAM BRINE INTERMEDIATE TANK & TRANSFER PUMP		
측 SCALE		
NONE		
일 DATE		
2025. 08. 14		
PROJECT NO.		REV.
223300		
도면번호 SHEET NO.		1A
CJ-W15N14-PR-220-000504		

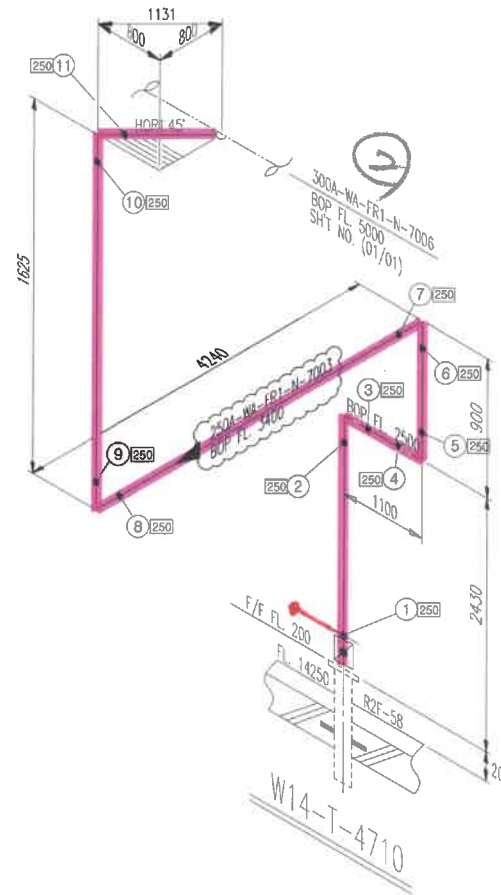


<NOTE>
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED TEE : TR * STUB-IN : SIU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : SIU * SOCKET(HDPE/PVC) : SG * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA : VG/VG1/VG2 * BALL VA : VB/VB1/VB2/VB2LO * BUTTERFLY VA : VF/VF4 * CHECK VA : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA : VL/VL1/VL2 * DIAPHRAGM VA : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINE) : FLXJ * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(W-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTD : LOTD * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHIELD : SH * SPECIAL BOLT & NUT : RI/RI1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	UNIT	R2FA	REF. PLAN DWG.	CJ-M15W14-PI-A30-020016-C	REF. P & ID	CJ-M15W14-PR-A20-000502,000501	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D
	THICK.	-	DESIGN	6.4	60										25.01.21	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
			OPERATING	4	AMB														

CLIENT	SK hynix	2
CONTRACTOR	SK ecoplant	
COMMODITY	WA	
LINE NO.	300A-WA-FR1-N-7006	
PROJECT NAME	M15X Ph4 Project	
JOB NO.	22330D	
SH'T NO.	01 / 01	REV. 0



1. TRAY & ANGLE, CHANNEL SUPPORT기 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

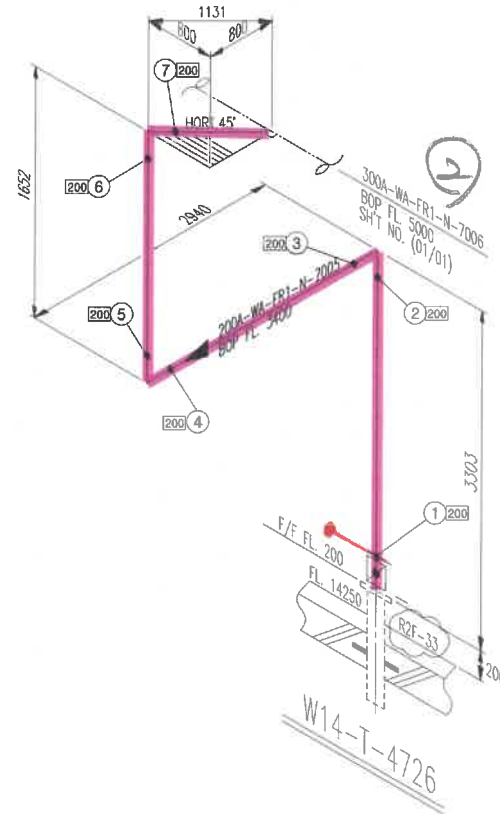
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA						
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020016-A							
						REF. P & ID CJ-M15W14-PR-A20-000501							
								25.03.31	APPROVED FOR CONSTRUCTION	황래권	인남후	박동원	
								DATE	DESCRIPTION	BY	CHK	APP	

ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	250A	11.4M	PIPE, FRP(MANUFACTURER STD), PE/BUT-JOINT, 4.8T
E90	250A	5EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 4.8T
F	250A	1EA	FLG, KS 10K, FT, FRP(MANUFACTURER STD), BUTT-JOINT
G	250A	2EA	GASKET, KS 10K, FT, EPDM, 3.0MM THK
INC	251K250A	11.4M	# 251K250A(1) + 250A(1) CASING
VP	250A	1EA	DAMPER, KS 10K, FT, BODY-FRP / DRING-EPDM, LEVER
B	250A	2SET	MACHINE BOLT & NUTS S15 304 / W WASHERS



SH'T NO.	REV.
01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유포할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.



1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



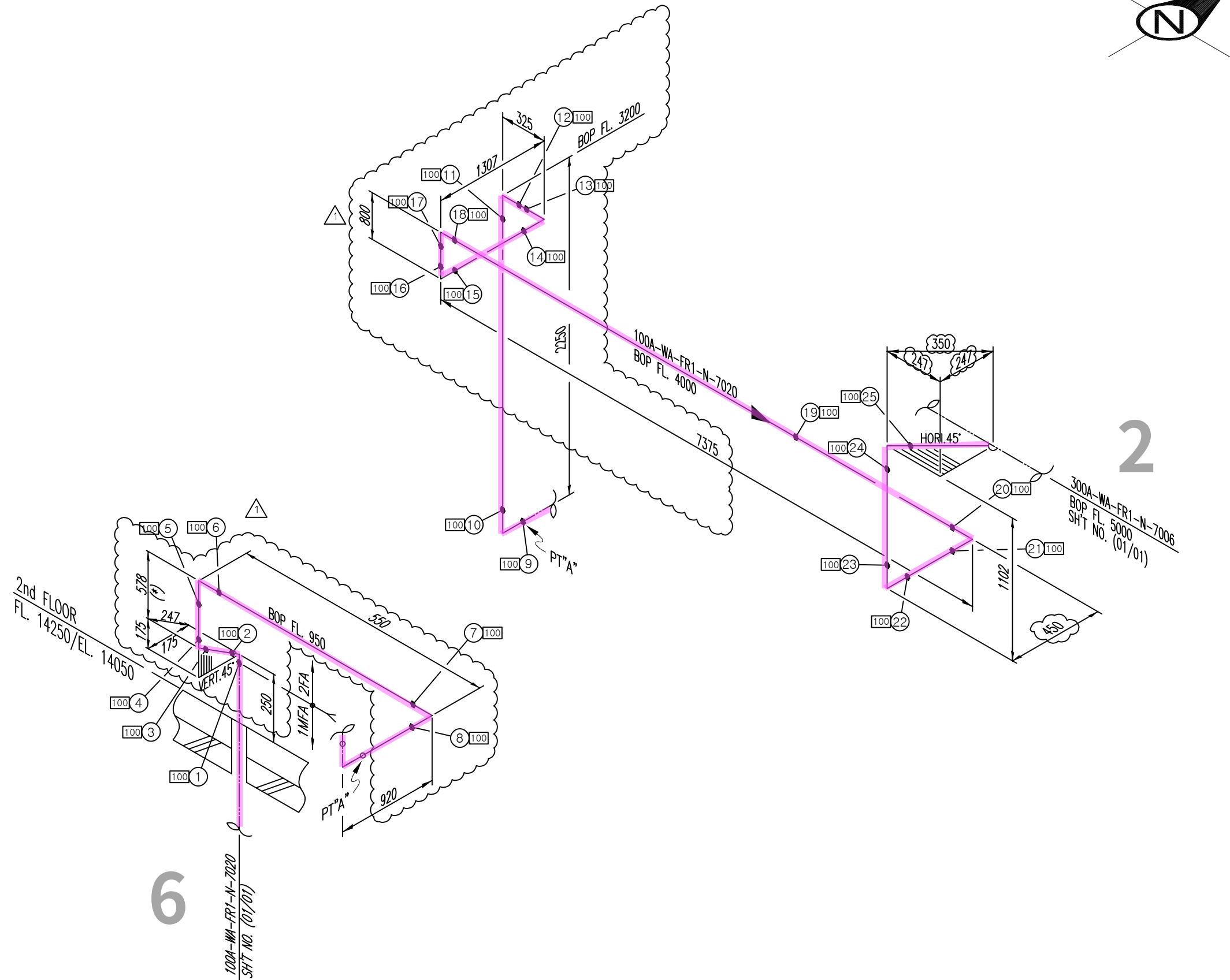
ITEM	SIZE	Q'TY	DESCRIPTION
FR1			
P	200A	9.0M	PIPE, FRP(MANUFACTURER STD), PE/BUTT-JOINT, 3.2T
E90	200A	3EA	ELBOW 90 DEG, FRP(MANUFACTURER STD), BUTT-JOINT, 3.2T
F	200A	1EA	FLG. KS 10K, FF, FRP(MANUFACTURER STD), BUTT-JOINT
VP	200A	1EA	DAMPER, KS 10K, FF, BODY-FRP / DRING-EPDM, LEVER
G	200A	2EA	GASKET, KS 10K, FF, EPDM, 3.0MM TH'K
B	200A	2SET	MACHINE BOLT & NUTS-SIS 304 / W WASHERS



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CONTRACTOR		PROJECT NAME	
		M15X PH4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.	200A-WA-FR1-N-7005	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

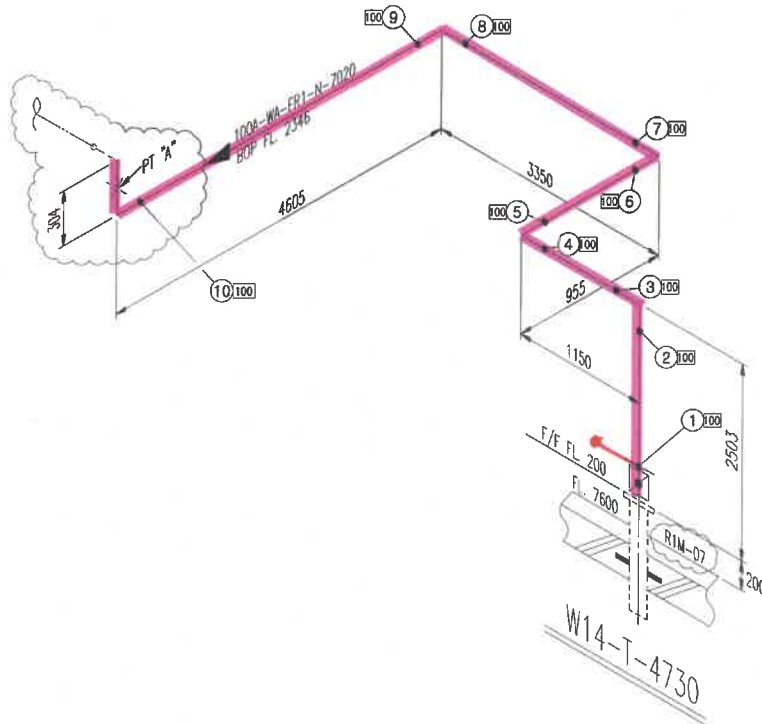


<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLX * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	△1 25.03.31	REVISED AS MARKED	황래중	안남후	박동영	CONTRACTOR	PROJECT NAME
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA						SK ecoplant	
			OPERATING	4	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-020015-A						COMMODITY	
						REF. P & ID	CJ-M15W14-PR-A20-000504	△0 25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	JOB NO.
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	22330D
													100A-WA-FR1-N-7020	SH'T NO. 01 / 01
														REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

CLIENT		 	
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
WA		22330D	
LINE NO.	SH# NO.	REV.	
100A-WA-FR1-N-7020	01 / 01	1	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유도자의 불법 사용으로 피해가 발생할 경우 유도자에게 민/형사상 책임을 요구할 수 있음.

[illegible]

* PIPE : P/P1	* SOT ELBOW : F90	* 45° ELBOW : F45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* COLLARED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* FLUGS : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET.HEMO.PAD : SC	* SADDLE.HEMO : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA : VG/GA/VG1	* BALL VA : VB/BA/VB1/VB2/AB2/B1	* BUTTERFLY VA : VF/VA1	* CHECK VA : VC/CA/VC2/AC2/BC2/AC2/BC2					
* GLOBE VA : VL/VL1/VL2	* DAPHNAS VA : VD/VD1	* DAMPER : DP	* QUICK COUPLING : SC	* FLEXIBLE JOINT(STEE) : FLXB		* ISOLATION JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL	* FLEXIBLE HOSE : FLXH							
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-YPE) : TSTR	* STRAINER(BLOCK-TYPE) : BSIR	* ANCHOR PLATE : STOP	* BARR SCREEN : BS		* EXHAUST QUIT : IXC	* LIN OADING BOSS : UBR	* AIR TRAP : AT	* FLOTTO : LOTO	* SIGHT GLASS : SG			* PRESSURE GAUGE : PG	* SHIELD : SH	* SPECIAL BOLT & NUT : EB/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14					CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	6.4	60	UNIT	R2FA						M15X Ph4 Project	
			OPERATING	4	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-020015-A						COMMODITY	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000502,000503							22330D	
						REV. DATE DESCRIPTION	BY	CHK	APP'D			LINE NO.	SHT NO.	REV.
						01 25.01.24 APPROVED FOR CONSTRUCTION	황래중	안남후	박동영			200A-WA-FR1-N-7004	01 / 01	0

경고: 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유표자의 불법 사용으로 피해가 발생할 경우 유표자에게 민/형사상 책임을 요구할 수 있음.

PUNCH LIST

Project No : 22330D

Package No :

Project Title : M15X Ph3 Project

Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Remarks

A : Before Hydro Test

B : Before M/C

C : Before Commissioning